Practical No. 9

Human-Computer Interaction (HCI) in Ubiquitous Computing

Aim

To develop an Android application that demonstrates the principles of Human-Computer Interaction (HCI) in the context of ubiquitous computing, by allowing a user to input data through a simple, intuitive interface and receive immediate, context-aware feedback.

Objectives

- To understand the basic principles of Human-Computer Interaction (HCI).
- To understand how ubiquitous computing enables seamless, everyday interaction between humans and computing devices.
- To design a simple and intuitive user interface using Android Studio.
- To implement event handling and immediate visual/textual feedback in response to user actions.
- To validate user input and handle basic error conditions gracefully.

Theory

Human-Computer Interaction (HCI) is a multidisciplinary field of study that focuses on the design, evaluation, and implementation of interactive computing systems, with an emphasis on making these systems usable, useful, and accessible to the people who use them. Good HCI design ensures that a user can interact with a system efficiently, effectively, and with a satisfying experience, through elements such as clear layouts, meaningful feedback, minimal cognitive load, and forgiving error handling.

Ubiquitous computing (also called pervasive computing) refers to the concept where computing is made to appear seamlessly available throughout the physical environment, embedded in everyday objects and activities, often invisibly, so that users can interact with technology naturally, without being conscious of the underlying computer. Smartphones are one of the most common examples of ubiquitous computing devices, since they allow people to interact with computing power anywhere, at any time, through simple touch-based interfaces.

This practical demonstrates HCI within a ubiquitous computing context by building a small, mobile-based application. The mobile device itself represents ubiquitous computing (an everyday, always-available device), while the app's simple form, immediate feedback, and input validation represent good HCI design practice: the user types their name, taps a button, and instantly receives a personalized, on-screen response, mimicking the immediate, low-friction interactions expected of pervasive computing systems.

Software Requirements

- Operating System: Windows / macOS / Linux
- Android Studio (Quail | 2024.3.1 or later)

- Android SDK (API level as configured by Android Studio)
- Java Development Kit (JDK)
- An Android Virtual Device (AVD) / Emulator or a physical Android device for testing

Procedure

1. Open Android Studio and create a new project using the "Empty Views Activity" template.
2. Name the application HCIApp and choose Java as the language.
3. Design the user interface in activity_main.xml using a vertical LinearLayout containing a title TextView, an EditText for name input, a Submit Button, and a result TextView.
4. Define the app's text resources (labels, hints, and messages) in strings.xml for clean separation of UI text from layout code.
5. In MainActivity.java, obtain references to the EditText, Button, and result TextView using findViewById().
6. Set an OnClickListener on the Submit button to capture the entered text when it is tapped.
7. Validate the input: if the name field is empty, display a Toast message asking the user to enter their name.
8. If the name field is not empty, update the result TextView with a personalized greeting and show a success Toast message.
9. Run the application on an emulator or a physical Android device and test both the empty-input and valid-input scenarios.

Code

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="24dp">

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Human-Computer Interaction"
        android:textSize="24sp"
        android:textStyle="bold"
        android:gravity="center"/>

    <EditText
        android:id="@+id/nameEditText"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="30dp"
        android:hint="Enter your name"/>

    <Button
        android:id="@+id/submitButton"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
```

```
        android:layout_marginTop="20dp"
        android:text="Submit"/>

    <TextView
        android:id="@+id/resultText"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="30dp"
        android:text="Your result will appear here"
        android:textSize="18sp"/>

</LinearLayout>
```

strings.xml

```
<resources>
    <string name="app_name">HCIApp</string>
    <string name="title">Human-Computer Interaction</string>
    <string name="name_hint">Enter your name</string>
    <string name="submit">Submit</string>
    <string name="result">Your result will appear here</string>
    <string name="empty_name">Please enter your name</string>
    <string name="success">Information submitted successfully!</string>
    <string name="hello_user">Hello %1$s!</string>
</resources>
```

MainActivity.java

```
package com.example.hciapp;

import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    EditText nameEditText;
    Button submitButton;
    TextView resultText;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        nameEditText = findViewById(R.id.nameEditText);
        submitButton = findViewById(R.id.submitButton);
        resultText = findViewById(R.id.resultText);

        submitButton.setOnClickListener(v -> {

            String name = nameEditText.getText().toString();
```

```
            if (name.isEmpty()) {
                Toast.makeText(this, "Please enter your name",
Toast.LENGTH_SHORT).show();
            } else {
                resultText.setText("Hello " + name + "!");
                Toast.makeText(this, "Information submitted successfully!",
Toast.LENGTH_SHORT).show();
            }
        });
    }
}
```

Output

On launching the app, the user is presented with the title "Human-Computer Interaction", a text field prompting for their name, and a Submit button, along with a placeholder result message.

- If Submit is tapped with the name field left empty, a Toast message "Please enter your name" appears, guiding the user to correct their input.
- If Submit is tapped after entering a name (e.g., "Riya"), the result TextView updates to display "Hello Riya!" and a Toast message "Information submitted successfully!" is shown.

Conclusion

Thus, an Android application demonstrating Human-Computer Interaction in ubiquitous computing was successfully designed and implemented. The application illustrates how a mobile device, as a ubiquitous computing platform, can provide a simple and intuitive interface, respond instantly to user actions, and gracefully handle invalid input, thereby reflecting core HCI principles of usability, feedback, and accessibility in everyday computing.